Deflating degrees of freedom: dirac16complex when x_5, x_6, x_7 deflate

A guide for a beginning student. Every equation and every number in this document was computed in the Mathematica notebook deflation.nb (17 input cells, evaluated without a single message), with exact arithmetic except for the one numerical integration of Section 8. All equations that are used are written out in this document.

1. The question

The coordinates are $x_0, x_1, \ldots, x_7$: x_0 is the hidden space direction, x_1, x_2, x_3 are ordinary 3-space, x_4 is time and x_5, x_6, x_7 are superluminal time directions. The flat metric is

$$\eta_{AB} = \mathrm{diag}(+1, +1, +1, +1, -1, -1, -1, -1).$$

We study the case in which x_5, x_6, x_7 are associated with metric components that each deflate exponentially in the time x_4:

$$g_{55} = b_{55}\, e^{-a x_4}, \qquad g_{66} = b_{66}\, e^{-a x_4}, \qquad g_{77} = b_{77}\, e^{-a x_4}, \qquad a > 0, \quad b_{55} > 0, \quad b_{66} > 0, \quad b_{77} > 0.$$

They enter the line element as $-g_{55}\, dx_5^2 - g_{66}\, dx_6^2 - g_{77}\, dx_7^2$, so x_5, x_6, x_7 remain timelike. They are called the **deflating degrees of freedom**. All other metric components are those of the primordial gravitational field, the vielbein of Pre-gravity6plane.nb with $a_4(Hx_4) = Hx_4$. With

$$z = 6Hx_0$$

the line element is

$$ds^2 = \tan^2 z\, dx_0^2 + e^{-2Hx_4} \sin^{-1/3} z\, \left(dx_1^2 + dx_2^2 + dx_3^2\right) - dx_4^2 - g_{55}\, dx_5^2 - g_{66}\, dx_6^2 - g_{77}\, dx_7^2,$$

on the domain $H > 0$, $0 < x_0 < \pi/(12H)$, that is $0 < z < \pi/2$, where $\tan z > 0$ and $\sin z > 0$.

Notation. g_{55}, g_{66}, g_{77} always denote the positive functions above. The metric tensor is written $g_{\mu\nu}$ with Greek indices; it is diagonal, and its entries for $\mu = 5, 6, 7$ are $-g_{55}, -g_{66}, -g_{77}$. The letter i runs over $1, 2, 3$ and j over $5, 6, 7$; capital letters $A, B, \ldots$ are frame indices $0, \ldots, 7$. Repeated indices are summed unless "no sum" is written.

2. Vielbein, volume element and the deflation of x_5, x_6, x_7

The vielbein is diagonal,

$$e^A{}_\mu = \mathrm{diag}(e_0, e_1, \ldots, e_7), \qquad g_{\mu\nu} = \eta_{AB}\, e^A{}_\mu\, e^B{}_\nu,$$

$$e_0 = \tan z, \qquad e_1 = e_2 = e_3 = e^{-Hx_4} \sin^{-1/6} z, \qquad e_4 = 1, \qquad e_j = \sqrt{g_{jj}} = \sqrt{b_{jj}}\, e^{-a x_4/2} \quad (j = 5, 6, 7),$$

and the inverse vielbein is $E_A{}^\mu = \delta^\mu_A / e_A$. The notebook confirms $e_j^2 = g_{jj}$ and

$$g_{\mu\nu} = \mathrm{diag}\Big(\tan^2 z,\ e^{-2Hx_4} \sin^{-1/3} z,\ e^{-2Hx_4} \sin^{-1/3} z,\ e^{-2Hx_4} \sin^{-1/3} z,\\ -1,\ -b_{55} e^{-a x_4},\ -b_{66} e^{-a x_4},\ -b_{77} e^{-a x_4}\Big).$$

Deflation. A coordinate interval Δx_j has the proper size

$$L_j = \sqrt{g_{jj}}\, \Delta x_j = \sqrt{b_{jj}}\, e^{-a x_4/2}\, \Delta x_j, \qquad \frac{d \ln L_j}{dx_4} = -\frac{a}{2},$$

so every proper size along x_5, x_6, x_7 shrinks by the factor e^{-1} in the time $\Delta x_4 = 2/a$. The volume factor of the deflating 3-space is

$$\sqrt{g_{55}\, g_{66}\, g_{77}} = \sqrt{b_{55} b_{66} b_{77}}\, e^{-3a x_4/2},$$

and the volume element of the eight-dimensional space, with $\left(\sqrt{|g|}\right)^2 = |\det g_{\mu\nu}|$, is

$$\sqrt{|g|} = e_0 e_1 \cdots e_7 = \frac{\sqrt{b_{55} b_{66} b_{77}\, \sin z}}{\cos z}\, e^{-3(a+2H)x_4/2} = \sqrt{b_{55} b_{66} b_{77}}\, \tan z\, \sin^{-1/2} z\, e^{-3(H+a/2)x_4}.$$

3. The Dirac matrices, the field dirac16complex and its two mirror sectors

The eight real 16x16 Dirac matrices are those of Pre-gravity6plane.nb. With $h = 1,2,3$, $p,q = 1,\ldots,4$ and the Levi-Civita symbol ϵ_{hpq4} ($\epsilon_{1234} = 1$):

$$\sigma = \begin{pmatrix} 0 & I_4 \\ I_4 & 0 \end{pmatrix}, \qquad (s_h)_{pq} = \epsilon_{hpq4} - \delta_{p4}\delta_{qh} + \delta_{ph}\delta_{q4}, \qquad (t_h)_{pq} = \epsilon_{hpq4} + \delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4},$$

$$\tau_0 = I_8, \quad \tau_h = \begin{pmatrix} 0 & s_h \\ s_h & 0 \end{pmatrix}, \quad \tau_4 = \begin{pmatrix} 0 & t_3 \\ -t_3 & 0 \end{pmatrix}, \quad \tau_5 = \begin{pmatrix} 0 & t_2 \\ -t_2 & 0 \end{pmatrix}, \quad \tau_6 = \begin{pmatrix} 0 & t_1 \\ -t_1 & 0 \end{pmatrix}, \quad \tau_7 = \tau_1\tau_2\tau_3\tau_4\tau_5\tau_6,$$

$$\bar\tau_A = \sigma\, \tau_A^T\, \sigma, \qquad T^A \equiv (T16^A)[A] = \begin{pmatrix} 0 & \bar\tau_A \\ \tau_A & 0 \end{pmatrix} \quad (A = 0,\ldots,7).$$

Proven exactly: all entries are -1, 0 or 1; $T^0,\ldots,T^3$ are symmetric and $T^4,\ldots,T^7$ antisymmetric;

$$T^A T^B + T^B T^A = 2\eta^{AB} I_{16}, \qquad \Gamma = T^0 T^1 \cdots T^7 = \mathrm{diag}(-I_8, I_8), \qquad \sigma_{16} = T^0 T^1 T^2 T^3 = \mathrm{diag}(-\sigma, \sigma),$$

$$(\sigma_{16} T^A)^T = -\sigma_{16} T^A, \qquad S^{AB} = \tfrac{1}{4}[T^A, T^B], \qquad [S^{AB}, T^C] = \eta^{BC} T^A - \eta^{AC} T^B.$$

σ_{16} is symmetric because σ is. The signature of the deflation field is still (4,4), so these remain the Dirac matrices of dirac16complex. By the last identity, the 28 matrices S^{AB} act on the T^C as the generators of SO(4,4) act on a vector.

The field. dirac16complex is the 16-component Pin(4,4) spinor field $\psi = (\psi_0,\ldots,\psi_{15})^T$ whose complex components are anticommuting Grassmann variables; it is a second-quantized fermion field. For Grassmann elements θ, ζ:

$$\psi_a \psi_b = -\psi_b \psi_a, \qquad \psi_a \psi_b^* = -\psi_b^* \psi_a, \qquad \psi_a^2 = 0, \qquad (\theta\zeta)^* = \zeta^* \theta^*, \qquad \bar\psi = \psi^\dagger \sigma_{16}.$$

A commuting (c-number) solution of the linear field equation is a mode function of the field; as in Pre-gravity6plane.nb, a mode function is called the wave function of a universe.

Krein operator and mirror sectors. Let

$$K = -i\sigma_{16} T^4, \qquad \beta = -iT^4, \qquad \Pi_\pm = \tfrac{1}{2}\,(I_{16} \pm K)\,, \qquad \psi_\pm = \Pi_\pm \psi.$$

Proven: $K^\dagger = K$, $K^2 = I_{16}$, the eigenvalues of K are $+1$ and -1 (eight times each), $KT^A = T^A K$ for $A = 0,\ldots,4$, $KT^A = -T^A K$ for $A = 5,6,7$, and $\sigma_{16} = K\beta$, equivalently $\sigma_{16} T^4 = iK$. For $A = 0,\ldots,4$

$$\Pi_+ \sigma_{16} T^A \Pi_- = 0, \qquad \Pi_+ \sigma_{16} T^A \Pi_+ = \Pi_+ \beta T^A \Pi_+, \qquad \Pi_- \sigma_{16} T^A \Pi_- = -\Pi_- \beta T^A \Pi_-,$$

and the same three identities hold without T^A. The Hermitian conjugate of the first identity, with $(\sigma_{16} T^A)^T = -\sigma_{16} T^A$, gives $\Pi_- \sigma_{16} T^A \Pi_+ = 0$ as well. Hence, for derivatives along $x_0,\ldots,x_4$,

$$\bar\psi\, T^A \partial_A \psi = \psi_+^\dagger \beta T^A \partial_A \psi_+ - \psi_-^\dagger \beta T^A \partial_A \psi_- \quad (A = 0,\ldots,4), \qquad \bar\psi\psi = \psi_+^\dagger \beta \psi_+ - \psi_-^\dagger \beta \psi_- :$$

there are no cross terms, and the $K = -1$ sector enters with the opposite sign. For a mode $\psi = e^{-i\omega x_4} u$ the canonical energy density is

$$\tfrac{1}{2}\left(\bar\psi\, T^4 \partial_4 \psi - (\partial_4 \bar\psi)\, T^4 \psi\right) = \omega\, \psi^\dagger K \psi,$$

so a mode of the $K = -1$ sector carries the opposite energy of the same mode of the $K = +1$ sector. The energy of a universe is called its mass: $K = +1$ is the **+mass sector** and $K = -1$ the **-mass sector**.

4. Curvature

The curvature is computed from the Christoffel symbols (a symbol Γ with three indices is a Christoffel symbol; Γ without indices is the chirality matrix of Section 3):

$$\Gamma^{\lambda}{}_{\mu\nu} = \tfrac{1}{2}\, g^{\lambda\rho}\left(\partial_\mu g_{\rho\nu} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}\right),$$

$$R^{\kappa}{}_{\lambda\mu\nu} = \partial_\mu \Gamma^{\kappa}{}_{\lambda\nu} - \partial_\nu \Gamma^{\kappa}{}_{\lambda\mu} + \Gamma^{\kappa}{}_{\mu\rho}\Gamma^{\rho}{}_{\lambda\nu} - \Gamma^{\kappa}{}_{\nu\rho}\Gamma^{\rho}{}_{\lambda\mu}, \qquad R_{\lambda\nu} = R^{\kappa}{}_{\lambda\kappa\nu}, \qquad R = g^{\lambda\nu}R_{\lambda\nu},$$

$$G_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2}R\, g_{\mu\nu}, \qquad R_{\kappa\lambda\mu\nu}R^{\kappa\lambda\mu\nu} = \sum_{\kappa,\lambda,\mu,\nu} \frac{g_{\kappa\kappa}}{g_{\lambda\lambda}\, g_{\mu\mu}\, g_{\nu\nu}} \left(R^{\kappa}{}_{\lambda\mu\nu}\right)^2 \quad \text{(diagonal metric)}.$$

4.1 The full deflation field

The Ricci scalar and the diagonal mixed components of the Einstein tensor are

$$R = \frac{3}{8}\left[3(a^2+3aH-32H^2) - 4(a^2+3aH+32H^2)\cos 2z + a(a+3H)\cos 4z\right]\csc^4 z,$$

$$G^{0}{}_{0} = -\frac{3}{16}\left[(a^2+3aH+2H^2)(3+\cos 4z) - 4(a^2+3aH+6H^2)\cos 2z\right]\csc^4 z,$$

$$G^{i}{}_{i} = -\frac{3}{16}\left[3a^2+6aH-64H^2 - 4(a^2+2aH+20H^2)\cos 2z + a(a+2H)\cos 4z\right]\csc^4 z \quad (i=1,2,3,\ \text{no sum}),$$

$$G^{4}{}_{4} = \frac{3}{4}\left(a^2+4H^2\right) - \frac{R}{2},$$

$$G^{j}{}_{j} = -\frac{3}{32}\left[3(a^2+4aH-64H^2) - 4(a^2+4aH+64H^2)\cos 2z + a(a+4H)\cos 4z\right]\csc^4 z \quad (j=5,6,7,\ \text{no sum}).$$

The only non-zero off-diagonal Ricci components are

$$R_{04} = R_{40} = -3H^2\cot z, \qquad G_{04} = R_{04} \quad (\text{because } g_{04}=0),$$

they do not contain a. The Kretschmann scalar is

$$R_{\kappa\lambda\mu\nu}R^{\kappa\lambda\mu\nu} = \frac{3}{2}\left[a^4 + 6a^2H^2 + 8H^4 + H^4\left(1245 + 108\cos 2z - \cos 4z\right)\cot^4 z\, \csc^4 z + 8H^4\cos^2 2z\, \csc^8 z\right].$$

Three consequences:

- R, every $G^{\mu}{}_{\mu}$, R_{04} and the Kretschmann scalar depend on x_0 only, not on x_4. Deflation therefore creates no curvature singularity at any finite time x_4.
- Near $x_0 = 0$, to leading order, $R = -\dfrac{7}{108\,H^2x_0^4}$ and $R_{\kappa\lambda\mu\nu}R^{\kappa\lambda\mu\nu} = \dfrac{85}{69984\,H^4x_0^8}$, independent of a: the curvature singularity of the primordial field at $x_0 = 0$ remains.
- The term $\frac{3}{2}a^4$ of the Kretschmann scalar is exactly the Kretschmann scalar of the pure deflation field below.

4.2 The pure deflation field: a hyperbolic 4-space of times

The pure deflation field keeps only the deflation,

$$ds^2 = dx_0^2 + dx_1^2 + dx_2^2 + dx_3^2 - dx_4^2 - g_{55}\,dx_5^2 - g_{66}\,dx_6^2 - g_{77}\,dx_7^2,$$

and isolates the effect of the deflating degrees of freedom. Its curvature invariants are constants:

$$R = 3a^2, \qquad R_{\kappa\lambda\mu\nu}R^{\kappa\lambda\mu\nu} = \frac{3a^4}{2}, \qquad G^{\mu}{}_{\mu} = -\frac{3a^2}{2}\ (\mu = 0,1,2,3), \qquad G^{\mu}{}_{\mu} = -\frac{3a^2}{4}\ (\mu = 4,5,6,7) \quad \text{(no sum)}.$$

Every Riemann component with an index in $\{0,1,2,3\}$ vanishes, so $x_0, \ldots, x_3$ span a flat $\mathbb{R}^4$. The block (x_4, x_5, x_6, x_7) with metric $-dx_4^2 - g_{55}dx_5^2 - g_{66}dx_6^2 - g_{77}dx_7^2$ is maximally symmetric:

$$R_{\kappa\lambda\mu\nu} = \frac{a^2}{4}\left(g_{\kappa\mu}g_{\lambda\nu} - g_{\kappa\nu}g_{\lambda\mu}\right), \qquad R_{\kappa\lambda\mu\nu} = g_{\kappa\rho}R^{\rho}{}_{\lambda\mu\nu}.$$

With the new coordinates

$$w = \frac{2}{a}\,e^{ax_4/2} > 0, \qquad y_j = \sqrt{b_{jj}}\,x_j \quad (j = 5,6,7),$$

the metric of the block becomes exactly

$$-dx_4^2 - g_{55}\,dx_5^2 - g_{66}\,dx_6^2 - g_{77}\,dx_7^2 = -\frac{4}{a^2}\,\frac{dw^2 + dy_5^2 + dy_6^2 + dy_7^2}{w^2}.$$

The right side is minus the Poincare upper-half-space metric $\frac{4}{a^2}(dw^2+dy_5^2+dy_6^2+dy_7^2)/w^2$ of the hyperbolic 4-space H^4 of radius $2/a$. **Time and the three deflating degrees of freedom together form a hyperbolic 4-space whose four directions are all timelike.** Its radius is $2/a$. The infinite past $x_4 \to -\infty$ is the boundary $w = 0$, the infinite future $x_4 \to +\infty$ is $w \to \infty$. This is why the invariants are constant: $R = 12/(2/a)^2 = 3a^2$ and $R_{\kappa\lambda\mu\nu}R^{\kappa\lambda\mu\nu} = 24/(2/a)^4 = 3a^4/2$.

5. Geodesics: deflation turns world lines back in time

Consider a world line $x^{\mu}(\ell)$ in the (x_4, x_5) plane of the deflation field, with affine parameter ℓ and $\dot{x} = dx/d\ell$. Nothing depends on x_5, so the momentum

$$p_5 = g_{5\mu}\,\dot{x}^{\mu} = -g_{55}\,\dot{x}_5$$

is conserved, and

$$\dot{x}_5 = -\frac{p_5}{b_{55}}\,e^{ax_4}$$

grows exponentially. The normalization $-\dot{x}_4^2 - g_{55}\,\dot{x}_5^2 = -1$ gives

$$\dot{x}_4^2 = 1 - \frac{p_5^2}{b_{55}}\,e^{ax_4}.$$

Since $\dot{x}_4^2 \geq 0$, the world line cannot go beyond the **turning point**

$$x_4^* = \frac{1}{a}\ln\frac{b_{55}}{p_5^2}.$$

The x_4 component of the geodesic equation, $\ddot{x}_4 = -\Gamma^4{}_{55}\,\dot{x}_5^2$ with $\ddot{x}_4 = \frac{1}{2}\,d(\dot{x}_4^2)/dx_4$, is satisfied. In the coordinates $w = (2/a)e^{ax_4/2}$, $y_5 = \sqrt{b_{55}}\,x_5$ of Section 4.2,

$$\left(\frac{dw}{dy_5}\right)^2 = \frac{w_*^2 - w^2}{w^2}, \qquad w_* = \frac{2\sqrt{b_{55}}}{a\,|p_5|},$$

whose solutions are the semicircles $(y_5 - y_{5c})^2 + w^2 = w_*^2$ standing on the boundary $w = 0$ (y_{5c} is a constant). The top of the semicircle, $w = w_*$, is exactly the turning point x_4^*. The world line comes from $x_4 \to -\infty$, rises to x_4^* and returns to $x_4 \to -\infty$: **deflation turns world lines back in time.**

6. The field equation of dirac16complex in the deflation field

The curved-space Dirac matrices and the spin connection are

$$\gamma^\mu = E_A{}^\mu\, T^A, \qquad D_\mu\psi = \partial_\mu\psi + \Omega_\mu\psi, \qquad D_\mu\bar\psi = \partial_\mu\bar\psi - \bar\psi\,\Omega_\mu,$$

$$\Omega_\mu = \tfrac{1}{4}\,\omega_\mu{}^A{}_B\, T_A T^B, \qquad T_A = \eta_{AB}T^B, \qquad \omega_\mu{}^A{}_B = e^A{}_\nu\left(\partial_\mu E_B{}^\nu + \Gamma^\nu{}_{\mu\lambda}E_B{}^\lambda\right).$$

The Lagrangian density of dirac16complex, with real constants m (mass) and λ (quartic self-coupling), is

$$\mathcal{L} = \sqrt{|g|}\left[\tfrac{1}{2}\left(\bar\psi\gamma^\mu D_\mu\psi - (D_\mu\bar\psi)\gamma^\mu\psi\right) - m\,\bar\psi\psi - \frac{\lambda}{2}\,(\bar\psi\psi)^2\right].$$

Treating ψ_a and ψ_a^* as independent Grassmann variables, the Euler-Lagrange equations with left derivatives,

$$\frac{\partial^L\mathcal{L}}{\partial\psi_a^*} - \partial_\mu\frac{\partial^L\mathcal{L}}{\partial(\partial_\mu\psi_a^*)} = 0 \quad (a = 0,\ldots,15),$$

were computed exactly in the Grassmann algebra; they equal

$$\sqrt{|g|}\,\sigma_{16}\left(\gamma^\mu D_\mu\psi - m\psi - \lambda(\bar\psi\psi)\psi\right) = 0.$$

Also proven: γ^μ is covariantly constant, $\partial_\mu\gamma^\nu + \Gamma^\nu{}_{\mu\lambda}\gamma^\lambda + [\Omega_\mu,\gamma^\nu] = 0$, and $\mathcal{L}$ is real, $\mathcal{L}^* = \mathcal{L}$. The inverse-vielbein factors and the spin-connection term are

$$\left(E_A{}^A\right)_{A=0..7} = \left(\cot z,\ e^{Hx_4}\sin^{1/6} z,\ e^{Hx_4}\sin^{1/6} z,\ e^{Hx_4}\sin^{1/6} z,\ 1,\ \frac{e^{ax_4/2}}{\sqrt{b_{55}}},\ \frac{e^{ax_4/2}}{\sqrt{b_{66}}},\ \frac{e^{ax_4/2}}{\sqrt{b_{77}}}\right),$$

$$\gamma^\mu\Omega_\mu = -\frac{3}{2}H\cot^2 z\, T^0 - \frac{3}{4}(a+2H)\,T^4,$$

so **the field equation of dirac16complex in the deflation field** is

$$\begin{aligned}\cot z\, T^0\partial_0\psi + e^{Hx_4}\sin^{1/6} z\sum_{i=1}^{3}T^i\partial_i\psi + T^4\partial_4\psi + e^{ax_4/2}\sum_{j=5}^{7}\frac{T^j\partial_j\psi}{\sqrt{b_{jj}}}\\ -\frac{3}{2}H\cot^2 z\, T^0\psi - \frac{3}{2}\left(H+\frac{a}{2}\right)T^4\psi = (m+\lambda\bar\psi\psi)\,\psi.\end{aligned}$$

The deflating directions enter through the growing factors $e^{ax_4/2}/\sqrt{b_{jj}}$, and deflation adds the friction term $-\frac{3a}{4}T^4\psi$.

Mass-sign pairing. Let $\mathcal{L}_m$ denote the Lagrangian with mass m. Define the map $\mathcal{C}\psi = \Gamma\psi^*$, i.e. $\psi_a \mapsto \Gamma_{ab}\psi_b^*$ and $\psi_a^* \mapsto \Gamma_{ab}\psi_b$, and the map $P_1\psi(x) = T^1\psi(R_1x)$ with the reflection $R_1 : x_1 \mapsto -x_1$. Proven exactly in the Grassmann algebra:

$$\mathcal{L}_m[\Gamma\psi^*] = \mathcal{L}_{-m}[\psi], \qquad \mathcal{L}_m[T^1\psi(R_1x)] = \mathcal{L}_{-m}[\psi](R_1x).$$

Each map sends every solution with mass $+m$ onto a solution with mass $-m$.

7. Universes: dispersion relation, turning point and critical deflation rate

Universes are mode functions: c-number solutions of the linear part of the field equation ($\lambda = 0$). In a mean-field (Kohn-Sham) treatment of the quartic term with a constant $\rho_s = \langle\bar\psi\psi\rangle$, m stands for the effective mass $m + \lambda\rho_s$.

Exact removal of the friction terms. With

$$\psi = F\chi, \qquad F = \sin^{1/4} z\; e^{3(a+2H)x_4/4}, \qquad \gamma^\mu\partial_\mu F + F\,\gamma^\mu\Omega_\mu = 0,$$

one has $\gamma^\mu D_\mu(F\chi) = F\gamma^\mu\partial_\mu\chi$, so the linear equation becomes $\gamma^\mu\partial_\mu\chi = m\chi$ without friction terms.

Dispersion relation. For a mode

$$\chi = \exp\left[i\left(k_0x_0 + k_1x_1 + k_2x_2 + k_3x_3 + q_5x_5 + q_6x_6 + q_7x_7 - \omega x_4\right)\right] u,$$

with frozen (adiabatic) coefficients $c_A = E_A{}^A = 1/e_A$, valid when the c_A change little over one wavelength, the equation becomes

$$(iS - i\omega T^4 - m)\, u = 0, \qquad S = \sum_{A\neq 4} c_A k_A\, T^A \quad (k_5, k_6, k_7 \equiv q_5, q_6, q_7).$$

Proven: $ST^4 + T^4S = 0$ and $S^2 = s^2 I_{16}$ with $s^2 = \sum_{A\neq 4} \eta_{AA}(c_Ak_A)^2$. With $X = iS - i\omega T^4$ and $(T^4)^2 = -I_{16}$ this gives $X^2 = -S^2 + \omega(ST^4 + T^4S) - \omega^2(T^4)^2 = (\omega^2 - s^2)I_{16}$ and $(X+m)(X-m) = (\omega^2 - s^2 - m^2)I_{16}$. Non-zero u therefore exist exactly when $\omega^2 = m^2 + s^2$:

$$\omega^2 = m^2 + \cot^2 z\, k_0^2 + e^{2Hx_4} \sin^{1/3} z\,(k_1^2 + k_2^2 + k_3^2) - e^{ax_4}\,\bar{q}^{\,2}, \qquad \bar{q}^{\,2} = \frac{q_5^2}{b_{55}} + \frac{q_6^2}{b_{66}} + \frac{q_7^2}{b_{77}}.$$

Momentum along a deflating (timelike) direction lowers ω^2, with the growing weight e^{ax_4}.

Turning point. For $k_1 = k_2 = k_3 = 0$ and $\bar{q}^{\,2} > 0$,

$$\omega^2 = m^2 + \cot^2 z\, k_0^2 - e^{ax_4}\,\bar{q}^{\,2}, \qquad \frac{d\omega^2}{dx_4} = -a\, e^{ax_4}\,\bar{q}^{\,2} < 0,$$

so ω^2 decreases strictly and vanishes at exactly one point,

$$x_4^* = \frac{1}{a} \ln \frac{m^2 + \cot^2 z\, k_0^2}{\bar{q}^{\,2}}.$$

Before x_4^* the universe oscillates ($\omega^2 > 0$); after it, the universe grows exponentially ($\omega^2 < 0$).

Critical deflation rate. With 3-space momentum the late-time sign of ω^2 is decided by e^{2Hx_4} against e^{ax_4}; the critical deflation rate is

$$a_c = 2H.$$

For $a > 2H$ and $\bar{q}^{\,2} > 0$, $\omega^2 \to -\infty$; for $a < 2H$ and $k_1^2 + k_2^2 + k_3^2 > 0$, $\omega^2 \to +\infty$. At $a = 2H$ exactly,

$$\omega^2 = m^2 + \cot^2 z\, k_0^2 + e^{2Hx_4}\left(\sin^{1/3} z\,(k_1^2 + k_2^2 + k_3^2) - \bar{q}^{\,2}\right),$$

and the late-time sign is the sign of the bracket. Exact limits for $H = 1$, $x_0 = 1/10$ (so $z = 3/5$), $m = 1$, $k_0 = 0$, $k_1 = 1$, $k_2 = k_3 = 0$, $q_5 = 1$, $q_6 = q_7 = 0$, $b_{55} = b_{66} = b_{77} = 1$: $\omega^2 \to -\infty$ for $a = 3$; $\omega^2 \to -\infty$ for $a = 2$, because $\sin^{1/3}(3/5) = 0.8265 < 1$; $\omega^2 \to +\infty$ for $a = 1$.

8. Exact evolution of a universe in the pure deflation field

In the pure deflation field of Section 4.2 ($e_0 = e_1 = e_2 = e_3 = e_4 = 1$, $e_j = \sqrt{b_{jj}}\, e^{-ax_4/2}$), γ^μ is covariantly constant and

$$\gamma^\mu \Omega_\mu = -\frac{3a}{4}\, T^4.$$

With

$$\psi = e^{3ax_4/4}\, e^{i(k_1x_1 + q_5x_5)}\, \chi(x_4)$$

one has exactly

$$\gamma^\mu D_\mu \psi - m\psi = e^{3ax_4/4}\, e^{i(k_1x_1+q_5x_5)}\, T^4\,(\chi' - M\chi), \qquad M = T^4\left(ik_1T^1 + iq_5\, c(x_4)\, T^5 - m\right), \qquad c(x_4) = \frac{e^{ax_4/2}}{\sqrt{b_{55}}},$$

so the linear field equation is the first-order system $\chi' = d\chi/dx_4 = M\chi$, with no approximation. With $c_0 = c_1 = c_2 = c_3 = 1$ the frozen-coefficient relation of Section 7 reads $\omega^2 = m^2 + k_1^2 - e^{ax_4}q_5^2/b_{55}$.

Numerical solution. Parameters $a = b_{55} = k_1 = m = 1$ and $q_5 = 1/2$, so

$$x_4^* = \frac{1}{a} \ln \frac{m^2 + k_1^2}{q_5^2 / b_{55}} = \ln 8 = 2.0794.$$

At $x_4 = -8$, where $c = e^{-4} = 0.018$, the universe starts as a unit eigenvector $\chi(-8)$ of

$$h_0 = -k_1 T^4 T^1 - imT^4 \qquad (\chi' = -ih_0\chi \text{ at } c = 0)$$

with eigenvalue $+\sqrt{k_1^2 + m^2} = \sqrt{2}$, lying in the $K = +1$ (+mass) sector; the notebook confirms $\chi^\dagger h_0 \chi = 1.4142135623730951$ and a $K = -1$ weight 0. The system was integrated with the Wolfram Language function NDSolve on $-8 \le x_4 \le 5$ (PrecisionGoal 10, AccuracyGoal 10, MaxSteps 10^6). With the norm $N = \chi^\dagger \chi$ and the sector weights $P_\pm = \chi^\dagger \Pi_\pm \chi$:

x_4	N	P_+	P_-	P_-/N	$P_+ - P_-$
-8	1	1	1.2326×10^{-31}	1.2326×10^{-31}	1
0	1.067409	1.033705	0.033705	0.031576	0.99999992
$x_4^* = \ln 8$	1.967250	1.483625	0.483625	0.245838	0.99999996
3	10.010161	5.505080	4.505080	0.450051	0.99999978
4	1198.190	not computed	not computed	not computed	not computed
5	1.058795×10^7	5.293977×10^6	5.293976×10^6	0.49999995	1.00000002

On the grid $x_4 = -8, -7, \ldots, 5$ the fraction P_-/N differs from $(N-1)/(2N)$ by at most 7.18×10^{-7}. Before the turning point the universe oscillates and its norm stays close to 1. After it, the norm grows exponentially, to 10^7 at $x_4 = 5$. The -mass weight P_- grows from 10^{-31} to 5.29×10^6, and the -mass fraction tends to 1/2. The notebook plots N (logarithmic scale, with a dashed line at x_4^*), P_-/N, and P_+ and P_- (logarithmic scale).

9. Pairs of +mass and -mass universes

(i) Mass-sign pairing. $\Gamma\psi^*$ and $T^1\psi(R_1 x)$ map solutions with mass $+m$ onto solutions with mass $-m$ (Section 6).

(ii) Frequency pairing. The linear field equation has real coefficients: T^A, ${E_A}^A$, the spin-connection coefficients and m are real. So the complex conjugate ψ^* of a solution is a solution, and it maps a mode of frequency ω and momenta k_A onto a mode of frequency $-\omega$ and momenta $-k_A$.

(iii) Mixing of the mirror sectors. The cross terms of direction A between the sectors are $(I_{16} + K)\,\sigma_{16}\,c_A T^A\,(I_{16} - K)$. They vanish for $A = 0, \ldots, 4$. For $A = 5, 6, 7$ their non-zero entries have the growing magnitude

$$\frac{2\,e^{ax_4/2}}{\sqrt{b_{jj}}}.$$

By Hermitian conjugation, $(I_{16} - K)\,\sigma_{16}\,c_A T^A\,(I_{16} + K)$ vanishes for the same A. Only the deflating directions couple the +mass and -mass sectors, and ever more strongly.

(iv) Exact pair creation: conservation of the Krein charge. For arbitrary real numbers κ_A $(A \ne 4)$ the matrix

$$M = T^4 \left(i \sum_{A \ne 4} \kappa_A T^A - m \right)$$

satisfies (proven)

$$M^\dagger K + KM = 0, \qquad M^\dagger + M = 2i\,T^4 \left(\kappa_5 T^5 + \kappa_6 T^6 + \kappa_7 T^7 \right), \qquad K \left(M^\dagger + M \right) + \left(M^\dagger + M \right) K = 0.$$

The equation of Section 8 is the case $\kappa_1 = k_1$, $\kappa_5 = q_5 c(x_4)$, all other $\kappa_A = 0$. Every frozen-coefficient mode of Section 7 obeys $\chi' = M\chi$ with $\kappa_A = c_A k_A$, because $T^4 \partial_4 \chi = (m - iS)\chi$ and $(T^4)^{-1} = -T^4$. For $\chi' = M\chi$,

$$\frac{d}{dx_4} \left(\chi^\dagger K \chi \right) = \chi^\dagger \left(M^\dagger K + KM \right) \chi = 0, \qquad \frac{d}{dx_4} \left(\chi^\dagger \chi \right) = \chi^\dagger \left(M^\dagger + M \right) \chi.$$

So the Krein charge

$$Q = \chi^\dagger K \chi = P_+ - P_-$$

is conserved. It is the x_4-component of the U(1) current of dirac16complex: $\bar{\psi}\gamma^4\psi = \psi^\dagger\sigma_{16}T^4\psi = i\,\psi^\dagger K\psi$ because $\gamma^4 = T^4$ and $\sigma_{16}T^4 = iK$; in the pure deflation field $\sqrt{|g|}\,\psi^\dagger K\psi = \sqrt{b_{55}b_{66}b_{77}}\;\chi^\dagger K\chi$. The norm $N = P_+ + P_-$ changes only through the deflating directions. Its generator $M^\dagger + M$ anticommutes with K, so it maps the +mass sector into the -mass sector and back, and it can only create +mass and -mass weight together. With Q fixed,

$$P_+ = \frac{N+Q}{2}, \qquad P_- = \frac{N-Q}{2}, \qquad \frac{dP_+}{dx_4} = \frac{dP_-}{dx_4} = \frac{1}{2}\,\frac{dN}{dx_4}\,:$$

the +mass and -mass weights grow by exactly the same amount. In the example of Section 8, $Q = 1$ (the computed values deviate from 1 by less than 2.2×10^{-7}), so $P_-/N = (N-1)/(2N) \to 1/2$. The two sectors carry opposite energies (Section 3), so equal growth of P_+ and P_- is the creation of pairs of universes of opposite mass. **Deflation creates {+mass, -mass} universes in pairs.**

10. Summary of results

1. The deflating degrees of freedom have proper sizes $\sqrt{b_{jj}}\,e^{-ax_4/2}$ per unit coordinate interval, so the deflation rate is $-a/2$ and the e-folding time is $2/a$. Their volume factor is $\sqrt{b_{55}b_{66}b_{77}}\,e^{-3ax_4/2}$, and the volume element is $\sqrt{|g|} = \sqrt{b_{55}b_{66}b_{77}}\tan z \sin^{-1/2} z\, e^{-3(H+a/2)x_4}$.

2. The curvature of the full field (R, $G^\mu{}_\nu$, Kretschmann scalar) depends on x_0 only; $R_{04} = -3H^2\cot z$ is the only off-diagonal Ricci component. There is no curvature singularity at finite x_4. Near $x_0 = 0$, $R = -7/(108H^2x_0^4)$ and $R_{\kappa\lambda\mu\nu}R^{\kappa\lambda\mu\nu} = 85/(69984H^4x_0^8)$, independent of a.

3. The pure deflation field is flat $\mathbb{R}^4$ times a hyperbolic 4-space H^4 of radius $2/a$ formed by x_4 and the deflating degrees of freedom, all four directions timelike: $R = 3a^2$, Kretschmann scalar $3a^4/2$, $G^\mu{}_\mu = -3a^2/2$ ($\mu = 0..3$) and $-3a^2/4$ ($\mu = 4..7$).

4. World lines with momentum p_5 along a deflating direction turn back at $x_4^* = (1/a)\ln(b_{55}/p_5^2)$; in H^4 they are semicircles of radius $w_* = 2\sqrt{b_{55}}/(a|p_5|)$.

5. The field equation of dirac16complex is derived exactly from the Grassmann Lagrangian: γ^μ is covariantly constant, $\mathcal{L}$ is real, mass-sign pairing holds, and $\psi = \sin^{1/4} z\, e^{3(a+2H)x_4/4}\chi$ removes the friction terms exactly.

6. The dispersion relation is $\omega^2 = m^2 + \cot^2 z\, k_0^2 + e^{2Hx_4}\sin^{1/3} z\,(k_1^2 + k_2^2 + k_3^2) - e^{ax_4}\bar{q}^{\,2}$. Universes turn from oscillation to exponential growth at $x_4^* = (1/a)\ln[(m^2 + \cot^2 z\, k_0^2)/\bar{q}^{\,2}]$. The critical deflation rate is $a_c = 2H$.

7. In the exact pure-deflation evolution, the norm grows from 1 to 1.06×10^7 between $x_4 = -8$ and $x_4 = 5$ ($x_4^* = \ln 8$), and the -mass fraction grows from 1.2×10^{-31} to 0.49999995.

8. The Krein charge $Q = P_+ - P_-$ is exactly conserved while deflation increases P_+ and P_- by equal amounts: deflation creates $\{+\text{mass}, -\text{mass}\}$ universes in pairs.

11. Assumptions and scope

1. The deflating components $g_{jj} = b_{jj}e^{-ax_4}$, their sign in the line element and the choice of the primordial field for all other components are those requested; the domain $0 < x_0 < \pi/(12H)$ is where the primordial vielbein is real and positive.

2. "Universe" means a mode function (c-number solution) of dirac16complex, as in Pre-gravity6plane.nb. Its "mass" is its energy for translations of x_4, or the mass parameter m in the mass-sign theorem.

3. The dispersion relation of Section 7 uses frozen (adiabatic) coefficients. The identities of Sections 2 to 6, the removal of the friction terms, the equation $\chi' = M\chi$ of Section 8 and the conservation of the Krein charge are exact. The numbers of Section 8 come from a numerical integration with the stated tolerances.

4. All theorems are exact statements about this model; they are not a proof about the physical universe.